

A Code-Based Approach to Secure Aggregation in Federated Learning

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Abstract—Federated learning (FL) trains shared models without centralizing data, yet the model updates clients exchange leak private information, and the secure-aggregation protocols that prevent this rely on key establishment that a quantum computer would break, exposing recorded updates under “harvest-now, decrypt-later.” Post-quantum FL secures this layer almost exclusively with *lattice* cryptography, concentrating an entire ecosystem on one hardness assumption that a single cryptanalytic advance would defeat. We propose CB-SAFE, a KEM-agnostic secure-aggregation framework whose post-quantum confidentiality is established by the NIST-standardized code-based KEM HQC rather than a lattice KEM, providing the same NIST security levels under an independent (syndrome-decoding) assumption. CB-SAFE exposes the KEM as a pluggable module (Definition 1), runs code-based HQC or lattice Kyber/ML-KEM under one masking protocol, and applies cross-round key amortization, ciphertext batching, and compression so the larger code-based payloads stay practical. A prototype liboqs implementation establishes initial feasibility, and we outline an equal-security evaluation against the lattice baseline across three public datasets (FEMNIST, CIFAR-10, Edge-IIoTset) and five metric families. To our knowledge, this is the first FL secure-aggregation framework built on a code-based KEM; the implementation will be released open-source.

Index Terms—Post-quantum cryptography, federated learning, secure aggregation, HQC, code-based cryptography, cryptographic diversity.

I. INTRODUCTION

Federated learning (FL) lets many clients train a shared model under a coordinating server while keeping their data local, but the transmitted model updates are not private: gradient-inversion and membership-inference attacks reconstruct training samples from them. Secure aggregation removes this leakage by ensuring the server observes only the sum of updates, through pairwise additive masks that cancel on aggregation. The confidentiality of these masks rests on classical key exchange, which Shor’s algorithm renders insecure once a cryptographically relevant quantum computer exists, exposing recorded updates retroactively under “harvest-now, decrypt-later.”

The natural defense replaces the key exchange with a post-quantum KEM, and recent post-quantum FL does this almost always with lattice cryptography: lattice multi-stage secret sharing [3], lattice key-homomorphic pseudo-random

functions [4], and lattice signature aggregation [5], with non-lattice work rare and code-free [6] and surveys confirming lattice dominance [7]. Concentrating confidentiality on one hardness assumption across an ecosystem is a systemic risk: a single cryptanalytic advance would break every such system at once.

NIST anticipated this risk when, in March 2025, it selected the code-based KEM HQC (whose security reduces to syndrome decoding of quasi-cyclic codes rather than to lattice problems) as a diversified backup to ML-KEM [1]. The same diversification belongs inside FL’s confidentiality layer; we realize it by making the KEM a pluggable module, so code-based and lattice key establishment run under one protocol and differ only in the swapped primitive. Our contributions are:

- **A KEM-agnostic secure-aggregation protocol** that formalizes masked aggregation over any IND-CCA2 KEM (Definition 1), making the post-quantum primitive a single replaceable module.
- **The first code-based instantiation, scope-qualified:** to our knowledge, CB-SAFE is the first FL secure-aggregation framework whose post-quantum confidentiality is established by the NIST-standardized code-based KEM (HQC) rather than a lattice KEM, at the same NIST security levels.
- **Ciphertext-handling for practicality:** cross-round key amortization, ciphertext batching, and compression that keep the code-based payloads practical inside the FL loop.
- **A planned reproducible implementation** (liboqs, Flower, PyTorch), prototyped to establish feasibility, with an equal-security evaluation across three datasets and five metric families (Section IV).

II. RELATED WORK

Post-quantum FL secures aggregation almost exclusively with lattices: multi-stage secret sharing [3], key-homomorphic PRFs [4], and signature aggregation [5], each advancing efficiency within the lattice family but inheriting its single-assumption exposure; non-lattice work is scarce and code-free (e.g., multivariate cryptography with blockchain [6]), and surveys confirm lattice dominance [7].

Three commitments are common to the work above and motivate CB-SAFE’s departure. (i) Confidentiality rests on a single lattice assumption [3]–[5]; CB-SAFE establishes it from an independent code-based assumption selected by NIST for exactly this diversity [1]. (ii) The post-quantum primitive is

TABLE I
CONFIGURATIONS COMPARED UNDER THE KEM-AGNOSTIC CB-SAFE
PROTOCOL (DEFINITION 1), AT IDENTICAL NIST SECURITY LEVELS;
CB-SAFE ADDS THE CODE-BASED VARIANT ABSENT FROM PRIOR
POST-QUANTUM FL.

Configuration	Family	Assumption	PQ-safe	Level
Plain FedAvg	none	n/a	No	n/a
CB-SAFE / lattice	lattice	Module-LWE	Yes	1/3/5
CB-SAFE / code-based	code-based	Syndrome decoding	Yes	1/3/5

hard-wired into each protocol; CB-SAFE exposes it behind a KEM-agnostic interface (Definition 1) so the code-based and lattice variants differ only in one module. (iii) No prior framework brings a standardized code-based KEM into FL secure aggregation. To our knowledge, none satisfies any of these three.

III. PROPOSED FRAMEWORK: CB-SAFE

System and threat model. We consider cross-device/cross-silo FL with a server and N clients running FedAvg over non-IID partitions. The server is honest-but-curious and may collude with up to $t < N/2$ clients; we additionally model malicious clients submitting poisoned updates and clients dropping out mid-round. The goal is that no coalition below the threshold learns any individual update x_i , only the aggregate $\sum_i x_i$.

Definition 1 (KEM-agnostic masked aggregation): Let $\mathcal{K} = (\text{KeyGen}, \text{Encaps}, \text{Decaps})$ be an IND-CCA2 KEM and $H : \{0, 1\}^* \rightarrow \mathcal{R}$ a key-derivation function into the update space $\mathcal{R}$. For an ordered pair (i, j) , client i computes $(c_{ij}, ss_{ij}) = \text{Encaps}(pk_j)$ and the mask $m_{ij} = H(ss_{ij})$, and client j recovers $ss_{ij} = \text{Decaps}(sk_j, c_{ij})$. Client i submits $y_i = x_i + \sum_{j>i} m_{ij} - \sum_{j<i} m_{ij}$; the server returns $\sum_i y_i = \sum_i x_i$, since each m_{ij} appears once with each sign.

Instantiating $\mathcal{K}$ with HQC yields the code-based variant; with Kyber/ML-KEM [2], the lattice baseline (Table I). As the protocol uses $\mathcal{K}$ only through its IND-CCA2 interface, correctness and privacy are identical across instantiations. Dropout is handled by secret-sharing each client’s seed so the server can cancel the masks of non-responding clients.

Making HQC practical. Replacing the KEM is not a drop-in substitution: code-based KEMs use larger ciphertexts and key material than lattice KEMs, so naive per-round encapsulation would inflate communication. CB-SAFE keeps the code-based variant practical by amortizing key establishment across rounds (long-lived HQC public keys are reused and only masks refreshed, so the $O(N)$ ciphertext exchange becomes a setup cost) and by batching each client’s pairwise ciphertexts and compressing key material. HQC cannot sign and no code-based signature is standardized, so we confine the code-based primitive to confidentiality and, where authentication is needed, compose a standardized lattice signature (ML-DSA), with only the KEM layer code-based.

IV. EVALUATION PLAN AND METRICS

We compare three configurations, plain FedAvg, the lattice (Kyber/ML-KEM) baseline, and the code-based (HQC) CB-

SAFE variant (Table I), across three public datasets in two domains: FEMNIST and non-IID CIFAR-10 (image classification) and Edge-IIoTset (IoT intrusion detection) [5], [6], all at matched NIST levels (1/3/5). We report five metric families: (1) *communication*, bytes/round/client separating model payload from key and ciphertext overhead; (2) *computation*, keygen/encaps/decaps time and per-round wall-clock; (3) *utility*, accuracy vs. rounds, which must coincide with the plain baseline since correct masking preserves the aggregate (Definition 1); (4) *robustness*, accuracy under malicious/colluding-client fractions and dropout; and (5) *security level*. Utility and robustness use paired tests with multiple-comparison correction over repeated seeds, establishing the first code-based reference point for post-quantum secure aggregation in FL.

V. CONCLUSION

Post-quantum FL has equated “quantum-safe” with “lattice-based,” leaving privacy-critical deployments exposed to a single cryptanalytic failure. CB-SAFE reframes the post-quantum KEM as a replaceable module and supplies the first code-based occupant of that slot, bringing the cryptographic diversity for which NIST standardized HQC into federated learning at equal security levels. By exposing the KEM behind a KEM-agnostic interface, CB-SAFE makes cryptographic agility a configuration choice as the post-quantum landscape evolves.

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